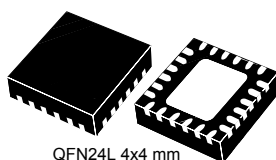


### Standalone USB PD sink controller with short-to-VBUS protections



QFN24L 4x4 mm

### Features

- Auto-run Type-C™ and USB PD sink controller
- Dead battery mode support
- Up to 3 sink PDO configurable profiles
- Dual high power charging path support
- Integrated  $V_{BUS}$  switch gate drivers (PMOS)
- Integrated  $V_{BUS}$  voltage monitoring
- Internal and/or external  $V_{BUS}$  discharge paths
- Short-to-VBUS protections on CC pins (22 V)
- High voltage capability on  $V_{BUS}$  pins (28 V)
- Dual power supply ( $V_{SYS}$  and/or  $V_{DD}$ ):
  - $V_{SYS} = [3.0 \text{ V}; 5.5 \text{ V}]$
  - $V_{DD} = [4.1 \text{ V}; 22 \text{ V}]$
- Debug accessory mode support
- Temperature range: -40 °C up to 105 °C
- ESD: 3 kV HBM - 1.5 kV CDM
- Certified:
  - USB Type-C™ rev 1.2
  - USB PD rev 2.0 (TID #1000133)
- Interoperable with USB PD rev 3.0

### Applications

- Set-top-box, hard disk drives, camcorders, cameras
- IoT, drones, accessories and battery powered devices
- LED lighting and industrial
- Healthcare and handheld devices
- Any Type-C sink device

### Description

The **STUSB4500** is a USB power delivery controller that addresses sink devices. It implements a proprietary algorithm to allow the negotiation of a power delivery contract with a source without MCU support (auto-run mode). PDO profiles are configured in an integrated non-volatile memory.

The device supports dead battery mode and is suited for sink devices powered from dead battery state and requiring high power charging profile to be fully operational.

Thanks to its 20 V technology, it implements high voltage features to protect the CC pins against short-circuits to  $V_{BUS}$  up to 22 V and to support high voltage on the  $V_{BUS}$  pins directly connected to the  $V_{BUS}$  power path up to 28 V.

| Product status link |                                                   |
|---------------------|---------------------------------------------------|
| STUSB4500           |                                                   |
| Device summary      |                                                   |
| Order code          | STUSB4500QTR                                      |
| Description         | Standalone USB PD sink controller (auto-run mode) |
| Package             | QFN24 EP 4x4 mm                                   |
| Marking             | 4500                                              |

## 1 Functional description

The STUSB4500 is a USB Type-C™ and power delivery controller IC for sink applications. It is able to negotiate a power delivery contract with a source without MCU support (auto-run mode). It relies on proprietary algorithms and configurable PDO (power data objects) thanks to an integrated non-volatile memory. It supports dead battery mode to allow a system to be powered from an external source directly. Combined with its capability to negotiate directly a power contract, the STUSB4500 is the ideal controller device for autonomous systems requiring high power charging profile to be fully operational.

**The STUSB4500 major role is to:**

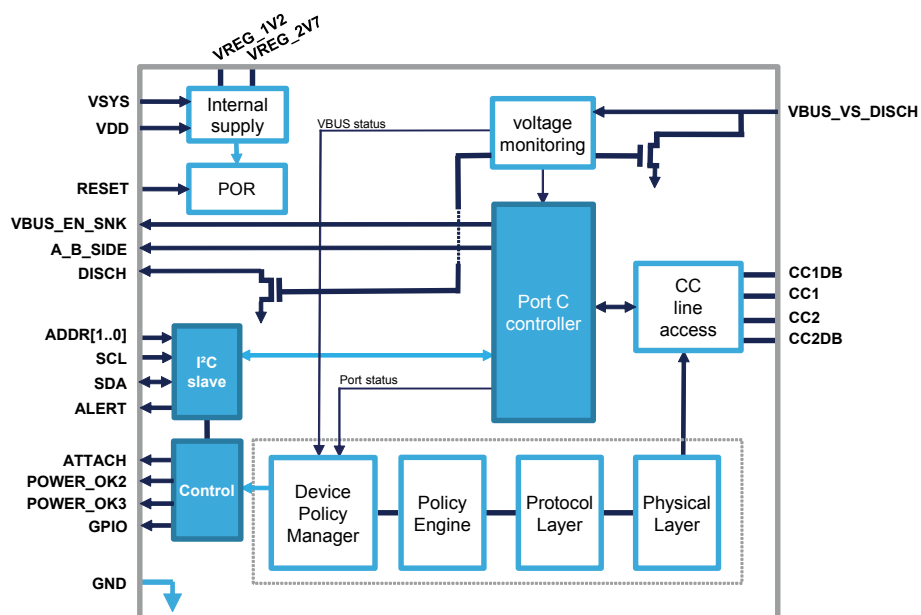
1. Detect the connection between two USB Type-C ports (attach detection)
2. Establish a valid source-to-sink connection
3. Determine the attached device mode: source or debug accessory
4. Resolve cable orientation and twist connections to establish USB data routing (MUX control)
5. Negotiate a USB power delivery (PD) contract with a PD capable source device
6. Configure the incoming  $V_{BUS}$  power path and the charging paths accordingly
7. Monitor the  $V_{BUS}$  power path and manage the  $V_{BUS}$  voltage transitions
8. Handle the high voltage protections

**The STUSB4500 also provides:**

- Dead battery mode
- PDO (power data object) customization through NVM
- Internal and/or external  $V_{BUS}$  discharge paths
- Dual high power charging path support
- Debug accessory mode detection
- Customization of the device configuration through NVM to support specific applications

### 1.1 Block overview

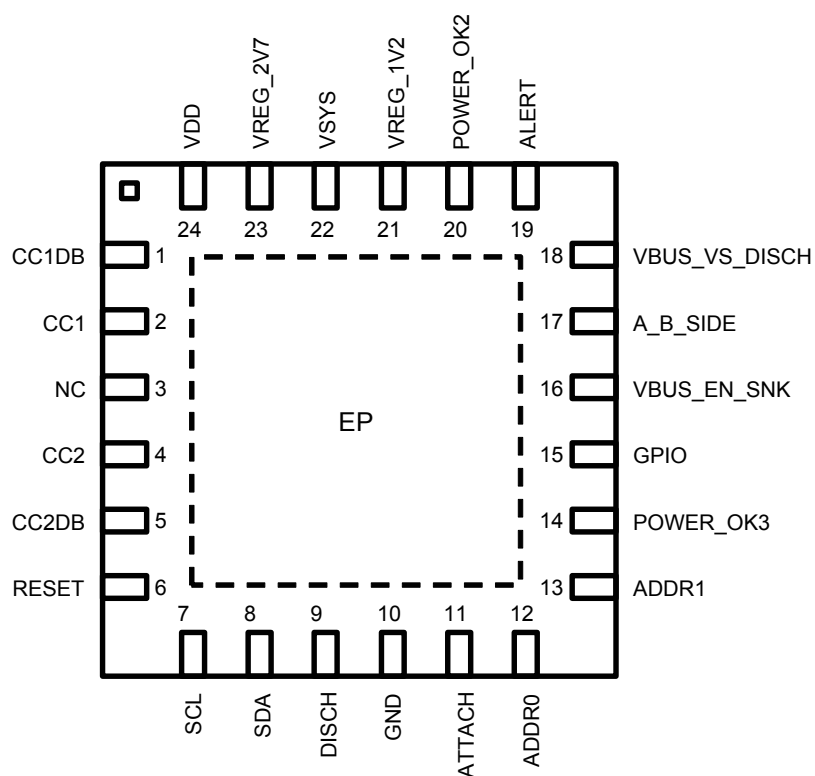
**Figure 1. Functional block diagram**



## 2 Inputs/outputs

### 2.1 Pinout

Figure 2. STUSB4500 pin connections (top view)



## 2.2 Pin list

**Table 1. Pin function list**

| Pin | Name          | Type     | Description                                                                      | Typical connection                                                                              |
|-----|---------------|----------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| 1   | CC1DB         | HV AIO   | Dead battery enable on CC1 pin                                                   | To CC1 pin if used or ground                                                                    |
| 2   | CC1           | HV AIO   | Type-C configuration channel 1                                                   | To Type-C receptacle A5                                                                         |
| 3   | NC            | -        | -                                                                                | Floating                                                                                        |
| 4   | CC2           | HV AIO   | Type-C configuration channel 2                                                   | To Type-C receptacle B5                                                                         |
| 5   | CC2DB         | HV AIO   | Dead battery enable on CC2 pin                                                   | To CC2 pin if used or ground                                                                    |
| 6   | RESET         | DI       | Reset input, active high                                                         | From system                                                                                     |
| 7   | SCL           | DI       | I <sup>2</sup> C clock input                                                     | To I <sup>2</sup> C master, ext. pull-up                                                        |
| 8   | SDA           | DI/OD    | I <sup>2</sup> C data input/output, active low open drain                        | To I <sup>2</sup> C master, ext. pull-up                                                        |
| 9   | DISCH         | HV AI/OD | Internal discharge path or external discharge path enable, active low open drain | From power system (internal path) or to the discharge path switch (external path), ext. pull-up |
| 10  | GND           | GND      | Ground                                                                           | Ground                                                                                          |
| 11  | ATTACH        | OD       | Attachment detection, active low open drain                                      | To MCU if any, ext. pull-up                                                                     |
| 12  | ADDR0         | DI       | I <sup>2</sup> C device address setting                                          | Static, to ground or ext. pull-up for address selection,<br>to ground if no connection to MCU   |
| 13  | ADDR1         | DI       | I <sup>2</sup> C device address setting                                          | Static, to ground or ext. pull-up for address selection,<br>to ground if no connection to MCU   |
| 14  | POWER_OK3     | OD       | Power contract flag, active low open drain                                       | To power system, ext. pull-up                                                                   |
| 15  | GPIO          | OD       | General purpose output, active low open drain                                    | To system, ext. pull-up                                                                         |
| 16  | VBUS_EN_SNK   | HV OD    | V <sub>BUS</sub> sink power path enable, active low open drain                   | To power switch or to power system, ext. pull-up                                                |
| 17  | A_B_SIDE      | OD       | Cable orientation, active low open drain                                         | USB super speed MUX select, ext. pull-up                                                        |
| 18  | VBUS_VS_DISCH | HV AI    | V <sub>BUS</sub> voltage monitoring and discharge path                           | From V <sub>BUS</sub> , receptacle side                                                         |
| 19  | ALERT         | OD       | I <sup>2</sup> C interrupt, active low open drain                                | To I <sup>2</sup> C master, ext. pull-up                                                        |
| 20  | POWER_OK2     | HV OD    | Power contract flag, active low open drain                                       | To power switch or to power system, ext. pull-up                                                |
| 21  | VREG_1V2      | PWR      | 1.2 V internal regulator output                                                  | 1 $\mu$ F typ. decoupling capacitor                                                             |
| 22  | VSYS          | PWR      | Power supply from system                                                         | From power system, connect to ground if not used                                                |
| 23  | VREG_2V7      | PWR      | 2.7 V internal regulator output                                                  | 1 $\mu$ F typ. decoupling capacitor                                                             |
| 24  | VDD           | HV PWR   | Power supply from USB power line                                                 | From V <sub>BUS</sub> , receptacle side                                                         |
| -   | EP            | GND      | Exposed pad is connected to ground                                               | To ground                                                                                       |

**Table 2. Pin function descriptions**

| Type | Description       |
|------|-------------------|
| D    | Digital           |
| A    | Analog            |
| O    | Output pad        |
| I    | Input pad         |
| IO   | Bidirectional pad |
| OD   | Open drain output |
| PD   | Pull-down         |
| PU   | Pull-up           |
| HV   | High voltage      |
| PWR  | Power             |
| GND  | Ground            |

## 2.3 Pin description

### 2.3.1 CC1/CC2

CC1 and CC2 are the configuration channel pins used for connection and attachment detection, plug orientation determination, USB power delivery communication, and system configuration management across USB Type-C cable. CC1 and CC2 are HiZ during reset.

### 2.3.2 CC1DB/CC2DB

CC1DB and CC2DB are used for dead battery mode. This mode is enabled by connecting CC1DB and CC2DB respectively to CC1 and CC2. Thanks to this connection, the pull-down terminations on the CC pins are present by default even if the device is not supplied (see [Section 3.5 Dead battery mode](#)).

Warning: CC1DB and CC2DB must be connected to ground when dead battery mode is not supported.

### 2.3.3 Reset

Active high reset.

### 2.3.4 I2C interface pins

**Table 3. I<sup>2</sup>C interface pin list**

| Name         | Description                                                                                    |
|--------------|------------------------------------------------------------------------------------------------|
| SCL          | I <sup>2</sup> C clock, need external pull-up                                                  |
| SDA          | I <sup>2</sup> C data, need external pull-up                                                   |
| ALERT        | I <sup>2</sup> C interrupt, need external pull-up                                              |
| ADDR0, ADDR1 | I <sup>2</sup> C device address bits (see <a href="#">Section 4 I<sup>2</sup>C Interface</a> ) |

Warning: ADDR0 and ADDR1 pins must be connected to ground when there is no connection to an MCU.

### 2.3.5 DISCH

This input/output pin can be used to implement a discharge path for highly capacitive  $V_{BUS}$  line on power system side.

When used as input, the discharge is internal and a serial resistor must be connected to the pin to limit the discharge current through the pin. Maximum discharge current is 500 mA.

The pin can be also used as an open drain output to control an external  $V_{BUS}$  discharge path when higher discharge current is required by the application, for instance.

The pin is activated at the same time as the internal discharge path on  $V_{BUS\_VS\_DISCH}$  pin. The discharge is activated automatically during cable disconnection, transition to a lower PDO voltage, hard reset and error recovery state. The discharge time is programmable by NVM (see [Section 5 Start-up configuration](#)).

### 2.3.6 GND

Ground.

### 2.3.7 ATTACH

This pin is asserted when a valid source-to-sink connection is established. It is also asserted when a connection to a debug accessory device is detected.

### 2.3.8 POWER\_OK2/POWER\_OK3

These pins report by default the status of the USB power delivery contract negotiation with the source.

Different configurations are proposed as stated in the table below to meet specific application requirements. The configuration of the POWER\_OK pins can be changed by NVM programming (see [Section 5 Start-up configuration](#)).

Depending on the programmed configuration, they can be used in combination with VBUS\_EN\_SNK pin to enable different power path scenarios.

POWER\_OK2 pin is a high voltage open drain output that allows a PMOS transistor to be directly driven to enable a power path.

POWER\_OK3 is a low voltage open drain output.

**Table 4. POWER\_OK pin configuration**

| Configuration ID<br>/NVM parameter<br>PWR_OK_CFG[1:0]                                                        | Pin name                   | Value | Description                                                                                  |
|--------------------------------------------------------------------------------------------------------------|----------------------------|-------|----------------------------------------------------------------------------------------------|
| Configuration 1: all PDOs on single VBUS power path                                                          |                            |       |                                                                                              |
| 00b                                                                                                          | VBUS_EN_SNK <sup>(1)</sup> | Hi-Z  | No source attached                                                                           |
|                                                                                                              |                            | 0     | Source attached                                                                              |
|                                                                                                              | POWER_OK2                  | Hi-Z  | No functionality                                                                             |
|                                                                                                              | POWER_OK3                  | Hi-Z  | No functionality                                                                             |
| Configuration 2: all PDOs on single VBUS power path + dedicated high power charging paths                    |                            |       |                                                                                              |
| 10b<br>(default)                                                                                             | VBUS_EN_SNK <sup>(1)</sup> | Hi-Z  | No source attached                                                                           |
|                                                                                                              |                            | 0     | Source attached                                                                              |
|                                                                                                              | POWER_OK2                  | Hi-Z  | No PD explicit contract                                                                      |
|                                                                                                              |                            | 0     | PD explicit contract with PDO2                                                               |
|                                                                                                              | POWER_OK3                  | Hi-Z  | No PD explicit contract                                                                      |
|                                                                                                              |                            | 0     | PD explicit contract with PDO3                                                               |
| Configuration 3: all PDOs on single VBUS power path + detection of USB Type-C current capability from source |                            |       |                                                                                              |
| 11b                                                                                                          | VBUS_EN_SNK <sup>(1)</sup> | Hi-Z  | No source attached                                                                           |
|                                                                                                              |                            | 0     | Source attached                                                                              |
|                                                                                                              | POWER_OK2                  | Hi-Z  | No source attached or source supplies default USB Type-C current at 5 V when source attached |
|                                                                                                              |                            | 0     | Source supplies 3.0 A USB Type-C current at 5 V when source attached                         |
|                                                                                                              | POWER_OK3                  | Hi-Z  | No source attached or source supplies default USB Type-C current at 5V when source attached  |
|                                                                                                              |                            | 0     | Source supplies 1.5 A USB Type-C current at 5 V when source attached                         |
| 01b                                                                                                          |                            |       | Not applicable                                                                               |

1. The VBUS\_EN\_SNK pin values correspond to the default behavior

In case of configuration 2 (default):

- When a PDO negotiation succeeds, the POWER\_OK pin related to the negotiated PDO is enabled (active low) when PS\_READY message is received from the source

- When a new PDO is negotiated upon source request, the active POWER\_OK pin is disabled (Hi-Z) when the STUSB4500 sends an RDO (request data object) message to the source with the new negotiated PDO
- At detachment the POWER\_OK pins remain enabled (if already asserted), whereas VBUS\_EN\_SNK is disabled (Hi-Z) to deactivate the V<sub>BUS</sub> power path from the USB Type-C receptacle. The POWER\_OK pins state is reinitialized (Hi-Z) after new attachment or after a reset

### 2.3.9 GPIO

This pin is an active low open drain output that can be configured by NVM as per table below (see [Section 5 Start-up configuration](#)).

**Table 5. GPIO pin configuration**

| NVM parameter<br>GPIO_CFG[1:0] | Pin name       | Pin function                                                                                                             | Value | Description                                                |
|--------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------|
| 00b                            | SW_CTRL_GPIO   | Software controlled GPIO.<br>The output state is defined by the value of I <sup>2</sup> C register bit #0 at address 2Dh | Hi-Z  | When bit #0 value is 0b (at start-up)                      |
|                                |                |                                                                                                                          | 0     | When bit #0 value is 1b                                    |
| 01b<br>(default)               | ERROR_RECOVERY | Hardware fault detection<br>(see <a href="#">Section 3.7 Hardware fault management</a> )                                 | Hi-Z  | No hardware fault detected                                 |
|                                |                |                                                                                                                          | 0     | Hardware fault detected                                    |
| 10b                            | DEBUG          | Debug accessory detection<br>(see <a href="#">Section 3.8 Debug accessory mode detection</a> )                           | Hi-Z  | No debug accessory detected                                |
|                                |                |                                                                                                                          | 0     | Debug accessory detected                                   |
| 11b                            | SINK_POWER     | Indicates USB Type-C current capability advertised by the source                                                         | Hi-Z  | Source supplies default or 1.5 A USB Type-C current at 5 V |
|                                |                |                                                                                                                          | 0     | Source supplies 3.0 A USB Type-C current at 5 V            |

### 2.3.10 VBUS\_EN\_SNK

This pin allows the incoming V<sub>BUS</sub> power from the USB Type-C receptacle to be enabled when a source is connected according to different operating conditions stated in the table below. The default behavior of the pin can be changed by NVM programming (see [Section 5 Start-up configuration](#)).

**Table 6. VBUS\_EN\_SNK pin configuration**

| NVM parameter<br>POWER_ONLY_ABOVE_5V | Pin function                                                                                                                  | Value | Description                                                     | Comment                                              |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------|------------------------------------------------------|
| 0b<br>(default)                      | Enables V <sub>BUS</sub> power path when source attached whatever V <sub>BUS</sub> voltage (5 V or any PDO voltage)           | Hi-Z  | No source attached                                              | Valid for all POWER_OK pin configurations 1, 2 and 3 |
|                                      |                                                                                                                               | 0     | Source attached                                                 |                                                      |
| 1b                                   | Enables V <sub>BUS</sub> power path only when source attached and V <sub>BUS</sub> voltage negotiated to PDO2 or PDO3 voltage | Hi-Z  | No source attached or no PD explicit contract with PDO2 or PDO3 |                                                      |
|                                      |                                                                                                                               | 0     | Source attached and PD explicit contract with PDO2 or PDO3      |                                                      |

When POWER\_ONLY\_ABOVE\_5V bit is set to logic level high, the VBUS\_EN\_SNK pin is asserted only when a PDO2 or PDO3 explicit contract is established with the source (see [Section 3.3 Auto-run mode](#)).

This feature is suited for sink devices requiring high power charging profile above 5 V to be fully operational (see [Section 6.1.2 Powering a system under high charging profile only](#)).

VBUS\_EN\_SNK pin is a high voltage open drain output that allows a PMOS transistor to be directly driven to enable the  $V_{BUS}$  power path.

### 2.3.11 A\_B\_SIDE

This output pin provides the cable orientation. It is used to establish USB SuperSpeed signal routing. This signal is not required in case of USB 2.0 support.

**Table 7. USB data MUX select**

| Value | Description                    |
|-------|--------------------------------|
| HiZ   | CC1 pin is attached to CC line |
| 0     | CC2 pin is attached to CC line |

### 2.3.12 VBUS\_VS\_DISCH

This input pin is used to sense  $V_{BUS}$  presence, monitor  $V_{BUS}$  voltage, and discharge  $V_{BUS}$  from the USB Type-C receptacle side.

A serial resistor connected to the pin must be used to limit the discharge current through the pin. Maximum discharge current is 50 mA.

The discharge is activated automatically during cable disconnection, transition to a lower PDO voltage, hard reset and error recovery state. The discharge time is programmable by NVM (see [Section 5 Start-up configuration](#)).

### 2.3.13 VREG\_1V2

This pin is used only for external decoupling of the 1.2 V internal regulator. The recommended decoupling capacitor is: 1  $\mu$ F typ. (0.5  $\mu$ F min., 10  $\mu$ F max.)

### 2.3.14 VSYS

This is the low power supply from the system, if there is any. It can be connected directly to a single cell Lithium battery or to the system power supply delivering 3.3 V or 5 V. It is recommended to connect the pin to ground when it is not used.

### 2.3.15 VREG\_2V7

This pin is used only for external decoupling of the 2.7 V internal regulator. The recommended decoupling capacitor is: 1  $\mu$ F typ. (0.5  $\mu$ F min., 10  $\mu$ F max.)

### 2.3.16 VDD

This is the power supply from the USB power line for applications powered by  $V_{BUS}$ .

## 3 Description of features

### 3.1 CC interface

The STUSB4500 controls the connection to the configuration channel (CC) pins, CC1 and CC2, through two main blocks: the CC line interface block and the CC control logic block.

The CC line interface block is used to:

- Set pull-down termination mode on the CC pins
- Monitor the CC pin voltage values related to the attachment detection thresholds
- Protect the CC pins against overvoltage

The CC control logic block is used to:

- Execute the Type-C FSM related to the sink power role with debug accessory support
- Determine the electrical state for each CC pin related to the detected thresholds
- Evaluate the conditions relative to the CC pin states and the  $V_{BUS}$  voltage value to transition from one state to another in the Type-C FSM
- Advertise a valid source-to-sink connection
- Determine the attached device mode: source or debug accessory
- Determine cable orientation to allow external routing of the USB data
- Manage USB Type-C power capability on  $V_{BUS}$ : USB default, medium or high current mode
- Handle hardware faults

### 3.2 Power delivery blocks

#### 3.2.1 Physical layer

The physical layer defines the signaling technology for USB power delivery. It is the physical link between CC pins and protocol layer. In Tx mode, it receives packet data from the protocol layer, calculates and appends a CRC, encodes the payload (i.e. packet data and CRC) and transmits the packet (i.e. preamble, SOP, payload, CRC and EOP) using biphase mark coding (i.e. BMC) over CC pins. In Rx mode, it recovers the clock and the data, detects the SOP, decodes the received data including the CRC, detects the EOP and validates the CRC.

#### 3.2.2 Protocol layer

The protocol layer has the responsibility to manage the messages from/to the physical layer. It automatically manages the protocol receive timeouts, the message counter, the retry counter and the GoodCRC messages. It communicates with the internal policy engine.

#### 3.2.3 Policy engine

The policy engine implements the power negotiation with the connected device according to its sink role. It implements all state machines controlling the protocol layer that forms and schedules the messages.

The policy engine uses the protocol layer to send/receive messages.

The policy engine interprets the device policy manager's input in order to implement policy for port and directs the protocol layer to send appropriate messages.

#### 3.2.4 Device policy manager

The device policy manager deals with the power capability request and change management. It operates according to the decision algorithm described in the following section.

### 3.3 Auto-run mode

The STUSB4500 implements a hardcoded decision algorithm that allows the device to negotiate in autonomous way a power delivery transaction with a source according to the PDO (power data objects) profiles programmed in the NVM.

It makes the STUSB4500 a plug-and-play, autonomous and effective solution to develop USB PD sink systems operating in standalone.

#### 3.3.1 Sink PDOs configuration

The STUSB4500 features up to 3 sink PDOs (SNK\_PDO). The value of each PDO is defined in the NVM (see [Section 5 Start-up configuration](#)).

**Table 8. Sink PDO description**

| Sink PDO # | Comment   | Priority | Description                                      |
|------------|-----------|----------|--------------------------------------------------|
| PDO1       | Mandatory | Low      | Defines the default power configuration          |
| PDO2       | Optional  | Medium   | Defines the intermediate power configuration     |
| PDO3       | Optional  | High     | Defines the highest power configuration (if any) |

PDO voltage configuration:

- PDO1 voltage is fixed to 5 V by hardware
- PDO2 and PDO3 voltages are programmable by NVM from 5 V to 20 V by steps of 50 mV as defined in the USB PD standard specification (see [Section 5 Start-up configuration](#))

PDO current configuration:

- The current of each PDO is programmable by NVM through look-up table see ([Section 5 Start-up configuration](#))
- 15 predefined values are set in the look-up table from 0.5 A to 5 A
- 1 custom value can be programmed in the look-up table from 10 mA to 5 A by steps of 10 mA as defined in the USB PD standard specification. This value is common to all PDOs if used

#### 3.3.2 Decision algorithm description

The decision algorithm compares each SNK\_PDO<sub>i</sub> with the SRC\_PDO<sub>j</sub> capabilities received from the source.

The comparison starts from the SNK\_PDO with the highest priority to the SNK\_PDO with the lowest priority. The voltage is compared first, the current afterwards.

A match occurs when both conditions are met:

1.  $V(\text{SNK\_PDO}_i) = V(\text{SRC\_PDO}_j)$
2.  $I(\text{SNK\_PDO}_i) \leq I(\text{SRC\_PDO}_j)$

The comparison loop stops at the first match. The remaining SRC\_PDO<sub>j</sub> are not compared and the SNK\_PDO<sub>i</sub> with lower priority are discarded.

In case of match:

- An RDO (request data object) message is formed with matched voltage  $V(\text{SNK\_PDO}_i)$  as operating voltage, related  $I(\text{SNK\_PDO}_i)$  current as operating current and  $I(\text{SRC\_PDO}_j)$  current from matched SRC\_PDO<sub>j</sub> as maximum current
- The RDO message is sent to the source for evaluation and acceptance by the source prior the transition to matched PDO voltage by the source and the reception of PS\_READY message by the sink

In case of no match:

- At the end of the comparison loop, if no match happens, the USB PD negotiation ends with an explicit USB PD contract at 5 V
- An RDO message is sent to the source with capability mismatch enabled, operating current set to current value from source PDO at 5 V, and maximum current set to  $I(\text{SNK\_PDO}_1)$

### 3.3.3 Requesting maximum source current

Thanks to dedicated NVM bit “REQ\_SRC\_CURRENT”, the operating current informed in the RDO message, when a matching PDO is found, can be set either to the current value from the matched sink PDO (default) or to the current value from the matched source PDO.

Requesting current value from the matched source PDO is useful for a sink that can benefit from higher power capability than originally required in order to increase its performance. This implies for the source to allocate a power reserve as stated in the USB PD standard specification.

In case the sink is not able to consume more power than requested, this option must not be used. It avoids allocating by the source a power reserve that is not used, thus limiting the overall power system optimization.

### 3.3.4 Decision algorithm application with examples

The following capabilities from the source have been considered to study the negotiation result for different sink PDO configuration cases with the STUSB4500:

- SRC\_PDO1 = 5 V, 3 A
- SRC\_PDO2 = 9 V, 3 A
- SRC\_PDO3 = 15 V, 2 A

**Table 9. Decision algorithm results for different cases**

| Case | Configured sink capabilities                                              | Result REQ_SRC_CURRENT = 0b                              | Result REQ_SRC_CURRENT = 1b                             |
|------|---------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| 1    | SNK_PDO2 = 9 V, 2.5 A<br>SNK_PDO1 = 5 V, 3 A                              | Match: RDO = 9 V, 2.5 A, 3 A<br>Not compared             | Match: RDO = 9 V, 3 A, 3 A<br>Not compared              |
| 2    | SNK_PDO3 = 9.1 V, 2.9 A<br>SNK_PDO2 = 8.9 V, 2.9 A<br>SNK_PDO1 = 5 V, 3 A | No match<br>No match<br>Match: RDO = 5 V, 3 A, 3 A       | No match<br>No match<br>Match: RDO = 5 V, 3 A, 3 A      |
| 3    | SNK_PDO3 = 15 V, 2.1 A<br>SNK_PDO2 = 9 V, 2.5 A<br>SNK_PDO1 = 5 V, 3 A    | No match<br>Match: RDO = 9 V, 2.5 A, 3 A<br>Not compared | No match<br>Match: RDO = 9 V, 3 A, 3 A<br>Not compared  |
| 4    | SNK_PDO3 = 15.1 V, 2 A<br>SNK_PDO2 = 15 V, 1 A<br>SNK_PDO1 = 5 V, 3 A     | No match<br>Match: RDO = 15 V, 1 A, 2 A<br>Not compared  | No match<br>Match: RDO = 15 V, 2 A, 2 A<br>Not compared |
| 5    | SNK_PDO2 = 15 V, 3 A<br>SNK_PDO3 = 9 V, 1 A<br>SNK_PDO1 = 5 V, 1 A        | No match<br>Match: RDO = 9 V, 1 A, 3 A<br>Not compared   | No match<br>Match: RDO = 9 V, 3 A, 3 A<br>Not compared  |

## 3.4 VBUS power path control

### 3.4.1 VBUS monitoring

The V<sub>BUS</sub> monitoring block supervises from the VBUS\_VS\_DISCH input pin the V<sub>BUS</sub> voltage on the USB Type-C receptacle side.

It is used to check that V<sub>BUS</sub> is within a valid voltage range to establish a valid source-to-sink connection and to enable safely the V<sub>BUS</sub> power path through the VBUS\_EN\_SNK pin.

It allows detection of unexpected V<sub>BUS</sub> voltage conditions such as undervoltage or overvoltage relative to the valid V<sub>BUS</sub> voltage range. When such conditions occur, the STUSB4500 reacts as follows:

- At attachment, it prevents the source-to-sink connection to be established and the V<sub>BUS</sub> power path to be asserted
- After attachment, it goes into unattached state and it disables the V<sub>BUS</sub> power path

The valid  $V_{BUS}$  voltage range is defined by a high limit  $V_{MONUSBH}$  and a low limit that can take as value either  $V_{THUSB}$  or  $V_{MONUSBL}$  depending on system operation and  $V_{BUS}$  voltage.

$V_{THUSB}$  low limit is fixed by hardware at 3.3 V. It corresponds to the undervoltage condition to detect a  $V_{BUS}$  disconnection when  $V_{BUS}$  voltage is at 5 V (USB Type-C or PDO1). The nominal value of  $V_{MONUSBL}$  is  $V_{BUS}-5\%$ . The low limit value can be shifted by fraction of  $V_{BUS}$  from -1% to -15%. The nominal value of  $V_{MONUSBH}$  is  $V_{BUS}+5\%$ . The high limit value can be shifted independently by fraction of  $V_{BUS}$  from +1% to +15%. It means the threshold limits can vary from  $V_{BUS}-5\%$  to  $V_{BUS}-20\%$  for the low limit and from  $V_{BUS}+5\%$  to  $V_{BUS}+20\%$  for the high limit.

At attachment, the valid  $V_{BUS}$  voltage range is defined by  $V_{MONUSBH}$  and  $V_{MONUSBL}$  limits to establish a valid source-to-sink connection. After attachment and during system operations, the valid  $V_{BUS}$  voltage range is automatically adjusted to  $V_{MONUSBH}$  and  $V_{THUSB}$  limits when  $V_{BUS}$  voltage is at 5 V (USB Type-C or PDO1), or to  $V_{MONUSBH}$  and  $V_{MONUSBL}$  limits when  $V_{BUS}$  operates under PDO2 or PDO3 voltage.

The  $V_{BUS}$  voltage value is automatically adjusted to 5 V (USB Type-C) at attachment and to the negotiated PDO voltage after PDO transition. During each PDO transition, the  $V_{BUS}$  monitoring is disabled for  $t_{SrcReady}$  (285 ms max.) as per USB PD standard specifications. Then the new limits applicable to the negotiated PDO voltage are monitored.

The threshold limits are preset by default in the NVM with different shift coefficients (see [Section 7.3 Electrical and timing characteristics](#)). The threshold limits can be changed independently through NVM programming (see [Section 5 Start-up configuration](#)).

### 3.4.2 VBUS discharge

The monitoring block also handles the  $V_{BUS}$  discharge paths connected to the  $VBUS\_VS\_DISCH$  pin for the USB Type-C receptacle side and to the  $DISCH$  pin for the power system side. The discharge paths are activated at the same time when disconnection is detected, during transition to a lower PDO voltage, when a hard reset is performed or when the device goes into the error recovery state (see [Section 3.7 Hardware fault management](#)). At detachment, during error recovery state or hard reset, the discharge is activated for  $T_{DISUSB0V}$  time. During transition to a lower PDO voltage, the discharge is activated for  $T_{DISUSBPDO}$  time.

The discharge time durations are also preset by default in the NVM (see [Section 7.3 Electrical and timing characteristics](#)). The discharge time durations can be changed through NVM programming (see [Section 5 Start-up configuration](#)).

The  $V_{BUS}$  discharge feature is enabled by default in the NVM and can be disabled through NVM programming (see [Section 5 Start-up configuration](#)).

### 3.4.3 VBUS power path assertion

The STUSB4500 can control the assertion of the  $V_{BUS}$  power path from the USB Type-C receptacle, directly or indirectly, through the  $VBUS\_EN\_SNK$  pin.

The table below summarizes the operating conditions that determine the electrical value of the  $VBUS\_EN\_SNK$  pin during system operation.

**Table 10. VBUS\_EN\_SNK pin behavior depending on the operating conditions**

| Value | Operating conditions                     |                      |                                      |                                                                                                 |                                           |
|-------|------------------------------------------|----------------------|--------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------|
|       | NVM parameter<br>POWER_ONLY<br>_ABOVE_5V | Connection<br>stage  | V <sub>BUS</sub> voltage from source | V <sub>BUS</sub> monitoring conditions on<br>VBUS_VS_DISCH pin                                  | Type-C state                              |
| 0     | 0b                                       | At attachment        | 5 V<br>(USB Type-C)                  | V <sub>BUS</sub> < V <sub>MONUSBH1</sub><br>and<br>V <sub>BUS</sub> > V <sub>MONUSBL1</sub>     | Attached.SNK<br>or<br>Debug Accessory.SNK |
|       |                                          | During operation     | 5 V<br>(USB Type-C<br>or SNK_PDO1)   | V <sub>BUS</sub> < V <sub>MONUSBH1</sub><br>and<br>V <sub>BUS</sub> > V <sub>THUSB</sub>        |                                           |
|       | 0b or 1b                                 | During operation     | V(SNK_PDO2)<br>or V(SNK_PDO3)        | V <sub>BUS</sub> < V <sub>MONUSBH2/3</sub><br>and<br>V <sub>BUS</sub> > V <sub>MONUSBL2/3</sub> |                                           |
| Hi-Z  | 0b or 1b                                 | Before<br>attachment | N.A.                                 | N.A.                                                                                            | Unattached.SNK                            |
|       | 1b                                       | At attachment        | 5 V<br>(USB Type-C)                  | V <sub>BUS</sub> < V <sub>MONUSBH1</sub><br>and<br>V <sub>BUS</sub> > V <sub>MONUSBL1</sub>     | Attached.SNK<br>or<br>Debug Accessory.SNK |
|       | 0b                                       |                      |                                      | V <sub>BUS</sub> > V <sub>MONUSBH1</sub><br>and<br>V <sub>BUS</sub> < V <sub>MONUSBL1</sub>     | AttachWait.SNK                            |
|       | 0b                                       | During operation     | 5 V<br>(USB Type-C<br>or SNK_PDO1)   | V <sub>BUS</sub> > V <sub>MONUSBH1</sub><br>or<br>V <sub>BUS</sub> < V <sub>THUSB</sub>         | Attached.SNK<br>or<br>Debug Accessory.SNK |
|       | 0b or 1b                                 |                      | V(SNK_PDO2)<br>or V(SNK_PDO3)        | V <sub>BUS</sub> > V <sub>MONUSBH2/3</sub><br>or<br>V <sub>BUS</sub> < V <sub>MONUSBL2/3</sub>  |                                           |

Type-C state column refers to the Type-C FSM states as defined in the USB Type-C standard specification.

### 3.5 Dead battery mode

Dead battery mode allows systems powered by a battery to be supplied by the V<sub>BUS</sub> when the battery is discharged and to start the battery charging process. This mode is also used in systems that are powered through the V<sub>BUS</sub> only.

Dead battery mode operates only if the CC1DB and CC2DB pins are connected respectively to the CC1 and CC2 pins. Thanks to these connections, the STUSB4500 presents a pull-down termination on its CC pins and advertises itself as a sink even if the device is not supplied.

When a source system connects to a USB Type-C port with the STUSB4500 configured in dead battery mode, it can detect the pull-down termination, establish the source-to-sink connection, and provide the V<sub>BUS</sub>. The STUSB4500 is then supplied thanks to the VDD pin connected to V<sub>BUS</sub> on the USB Type-C receptacle side. The STUSB4500 can finalize the connection on its side and enable the power path on V<sub>BUS</sub> thanks to the VBUS\_EN\_SNK pin to allow the system to be powered.

### 3.6 High voltage protections

The STUSB4500 can be safely used in systems or connected to systems that handle high voltage on the  $V_{BUS}$  power path. The device integrates an internal circuitry on the CC pins that tolerates high voltage and ensures protection up to 22 V in case of unexpected short-circuits with the  $V_{BUS}$  or in case of a connection to a device supplying high voltage on the  $V_{BUS}$ .

### 3.7 Hardware fault management

The STUSB4500 handles during system operation some pre-identified hardware fault conditions. When such conditions happen, the circuit goes into a transient error recovery state named ErrorRecovery in the Type-C FSM as defined in the USB Type-C standard specifications.

The error recovery state is equivalent to force a detach event. When entering in this state, the device de-asserts the  $V_{BUS}$  power path by disabling the  $VBUS\_EN\_SNK$ ,  $POWER\_OK2$  and  $POWER\_OK3$  pins, and it removes the terminations from the CC pins during several tens of milliseconds. Then, it transitions to the unattached state.

The STUSB4500 goes into error recovery state when at least one condition listed below is met:

- If an overtemperature is detected (junction temperature above maximum  $T_J$ )
- If an overvoltage is detected on the CC pins (voltage on CC pins above  $V_{OVP}$ )
- If after a hard reset the power delivery communication with the source is broken

The detection of a hardware fault is advertised through the GPIO pin when configured in `ERROR_RECOVERY` mode.

See [Section 7 Electrical characteristics](#) for threshold values.

### 3.8 Debug accessory mode detection

The STUSB4500 detects a connection to a debug and test system (DTS) as defined in the USB Type-C standard specification. The debug accessory detection is advertised through the GPIO pin when configured in `DEBUG` mode.

A debug accessory device is detected when both the CC1 and CC2 pins are pulled up by an  $R_p$  resistor from the connected device. The voltage levels on the CC1 and CC2 pins give the orientation and current capability as described in the table below. The GPIO pin configured in `DEBUG` mode is asserted to advertise the DTS detection and the `A_B_SIDE` pin indicates the orientation of the connection.

**Table 11. Orientation and current capability detection in sink power role**

| # | CC1 pin<br>(CC2 pin)   | CC2 pin<br>(CC1 pin)   | Charging current<br>configuration | A_B_SIDE pin<br>CC1/CC2<br>(CC2/CC1) |
|---|------------------------|------------------------|-----------------------------------|--------------------------------------|
| 1 | $R_p$ 3 A              | $R_p$ 1.5 A            | Default                           | Hi-Z (0)                             |
| 2 | $R_p$ 1.5 A            | $R_p$ default          | 1.5 A                             | Hi-Z (0)                             |
| 3 | $R_p$ 3 A              | $R_p$ default          | 3.0 A                             | Hi-Z (0)                             |
| 4 | $R_p$<br>def/1.5 A/3 A | $R_p$<br>def/1.5 A/3 A | Default                           | Hi-Z (Hi-Z)                          |

## 4 I<sup>2</sup>C interface

### 4.1 Read and write operations

The I<sup>2</sup>C interface is used to configure, control and read the operation status of the device. It is compatible with the Philips I<sup>2</sup>C Bus® (version 2.1). The I<sup>2</sup>C is a slave serial interface based on two signals:

- SCL - serial clock line: input clock used to shift data
- SDA - serial data line: input/output bidirectional data transfers

A filter rejects the potential spikes on the bus data line to preserve data integrity.

The bidirectional data line supports transfers up to 400 Kbit/s (fast mode). The data are shifted to and from the chip on the SDA line, MSB first.

The first bit must be high (START) followed by the 7-bit device address and the read/write control bit.

Four 7-bit device address are available for the STUSB4500 thanks to the external programming of DevADDR0 and DevADDR1 bits through ADDR0 and ADDR1 pins setting i.e. 0x28 or 0x29 or 0x2A or 0x2B. It allows four STUSB4500 devices to be connected on the same I<sup>2</sup>C bus.

**Table 12. Device address format**

| Bit7     | Bit6     | Bit5     | Bit4     | Bit3     | Bit2     | Bit1     | Bit0 |
|----------|----------|----------|----------|----------|----------|----------|------|
| DevADDR6 | DevADDR5 | DevADDR4 | DevADDR3 | DevADDR2 | DevADDR1 | DevADDR0 | R/W  |
| 0        | 1        | 0        | 1        | 0        | ADDR1    | ADDR0    | 0/1  |

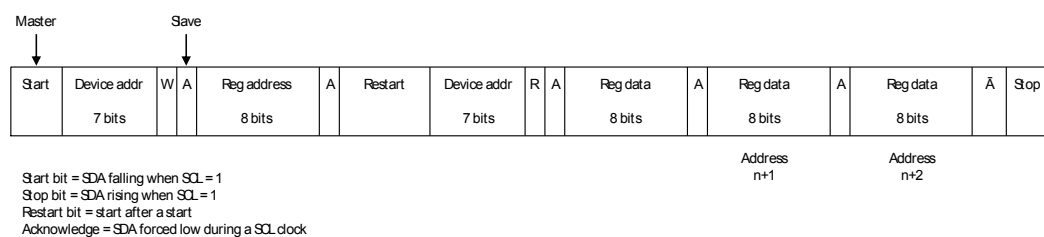
**Table 13. Register address format**

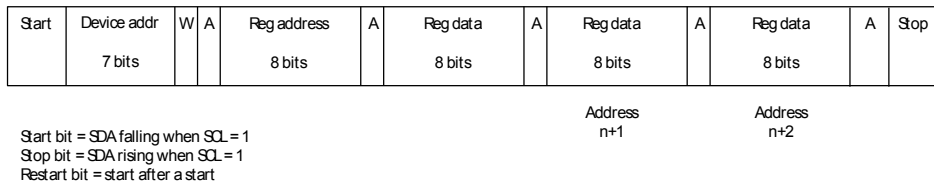
| Bit7     | Bit6     | Bit5     | Bit4     | Bit3     | Bit2     | Bit1     | Bit0     |
|----------|----------|----------|----------|----------|----------|----------|----------|
| RegADDR7 | RegADDR6 | RegADDR5 | RegADDR4 | RegADDR3 | RegADDR2 | RegADDR1 | RegADDR0 |

**Table 14. Register data format**

| Bit7  | Bit6  | Bit5  | Bit4  | Bit3  | Bit2  | Bit1  | Bit0  |
|-------|-------|-------|-------|-------|-------|-------|-------|
| DATA7 | DATA6 | DATA5 | DATA4 | DATA3 | DATA2 | DATA1 | DATA0 |

**Figure 3. Read operation**



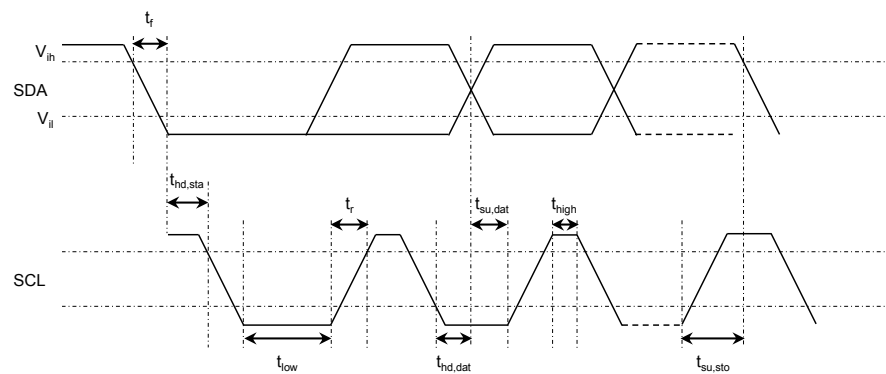
**Figure 4. Write operation**


## 4.2 Timing specifications

The device uses a standard slave I<sup>2</sup>C channel at speed up to 400 kHz.

**Table 15. I<sup>2</sup>C timing parameters - VDD = 5 V**

| Symbol              | Parameter                                        | Min.                    | Typ. | Max. | Unit |
|---------------------|--------------------------------------------------|-------------------------|------|------|------|
| F <sub>scl</sub>    | SCL clock frequency                              | 0                       |      | 400  | kHz  |
| t <sub>hd,sta</sub> | Hold time (repeated) START condition             | 0.6                     |      | -    | μs   |
| t <sub>low</sub>    | LOW period of the SCL clock                      | 1.3                     |      | -    |      |
| t <sub>high</sub>   | HIGH period of the SCL clock                     | 0.6                     |      | -    |      |
| t <sub>su,dat</sub> | Setup time for repeated START condition          | 0.6                     |      | -    |      |
| t <sub>hd,dat</sub> | Data hold time                                   | 0.04                    |      | 0.9  |      |
| t <sub>su,dat</sub> | Data setup time                                  | 100                     | -    | -    | ns   |
| t <sub>r</sub>      | Rise time of both SDA and SCL signals            | 20 + 0.1 C <sub>b</sub> |      | 300  |      |
| t <sub>f</sub>      | Fall time of both SDA and SCL signals            | 20 + 0.1 C <sub>b</sub> |      | 300  | μs   |
| t <sub>su,sto</sub> | Set-up time for STOP condition                   | 0.6                     |      | -    |      |
| t <sub>buf</sub>    | Bus free time between a STOP and START condition | 1.3                     |      | -    | pF   |
| C <sub>b</sub>      | Capacitive load for each bus line                | -                       |      | 400  |      |

**Figure 5. I<sup>2</sup>C timing diagram**


## 5 Start-up configuration

### 5.1 User-defined parameters

The STUSB4500 has a set of user-defined parameters that can be customized by NVM re-programming through the I<sup>2</sup>C interface. This feature allows the customer to change the preset configuration of the USB Type-C and PD interface and to define a new configuration to meet specific application requirements addressing various use cases, or specific implementations.

The NVM re-programming overrides the initial default setting to define a new default setting that is used at power-up or after a reset. The default setting is copied at power-up, or after a reset, from the embedded NVM into I<sup>2</sup>C registers. The values copied in the I<sup>2</sup>C registers are used by the STUSB4500 during the system operation.

The NVM re-programming is possible with a customer password. The I<sup>2</sup>C registers must be re-initialized after each NVM re-programming to make effective the new parameters setting either through power-off and power-up sequence, or through reset.

### 5.2 Default start-up configuration

The table below lists the user-defined parameters and indicates the default start-up configuration of the STUSB4500.

**Table 16. STUSB4500 user-defined parameters and default settings**

| Parameter name    | Parameter description      | Reset value (default) | Value                      | Description                                                                                                                                                            |
|-------------------|----------------------------|-----------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SNK_PDO_NUMB[1:0] | Number of sink PDOs        | 11b<br>(3)            | 00b                        | 1 PDO                                                                                                                                                                  |
|                   |                            |                       | 01b                        | 1 PDO                                                                                                                                                                  |
|                   |                            |                       | 10b                        | 2 PDOs                                                                                                                                                                 |
|                   |                            |                       | 11b                        | 3 PDOs                                                                                                                                                                 |
| V_SNK_PDO2        | Voltage value for SNK_PDO2 | 0.05*300<br>(15 V)    | 0.05*100<br>to<br>0.05*400 | Flexible voltage value<br>$5\text{ V} \leq 0.05 \cdot V_{\text{SNK\_PDO2\_FLEX}}[9:0] \leq 20\text{ V}$ by steps of 50 mV<br>Default V_SNK_PDO2_FLEX[9:0] = 0100101100 |
| V_SNK_PDO3        | Voltage value for SNK_PDO3 | 0.05*400<br>(20 V)    | 0.05*100<br>to<br>0.05*400 | Flexible voltage value<br>$5\text{ V} \leq 0.05 \cdot V_{\text{SNK\_PDO3\_FLEX}}[9:0] \leq 20\text{ V}$ by steps of 50 mV<br>Default V_SNK_PDO3_FLEX[9:0] = 0110010000 |
| I_SNK_PDO1        | Current value for SNK_PDO1 | 0101b<br>(1.5 A)      | 0000b                      | I_SNK_PDO_LUT[3:0] flexible current value from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                         |
|                   |                            |                       | 0001b<br>to<br>1111b       | I_SNK_PDO_LUT[3:0] pre-defined current values from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                     |

| Parameter name        | Parameter description                                                                                       | Reset value (default) | Value              | Description                                                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------|-----------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I_SNK_PDO2            | Current value for SNK_PDO2                                                                                  | 0101b<br>(1.5 A)      | 0000b              | I_SNK_PDO_LUT[3:0] flexible current value from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                                                                                                          |
|                       |                                                                                                             |                       | 0001b to 1111b     | I_SNK_PDO_LUT[3:0] pre-defined current values from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                                                                                                      |
| I_SNK_PDO3            | Current value for SNK_PDO3                                                                                  | 0011b<br>(1.0 A)      | 0000b              | I_SNK_PDO_LUT[3:0] flexible current value from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                                                                                                          |
|                       |                                                                                                             |                       | 0001b to 1111b     | I_SNK_PDO_LUT[3:0] pre-defined current values from look-up table (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> )                                                                                                      |
| I_SNK_PDO_FLEX        | Flexible current value common to all PDOs                                                                   | 0.01*200<br>(2.0 A)   | 0.01*1 to 0.01*500 | Flexible current value<br>$10\text{ mA} \leq 0.01 \cdot I_{\text{SNK\_PDO\_FLEX}}[9:0] \leq 5\text{ A}$ by steps of 10 mA<br>Default I_SNK_PDO_FLEX[9:0] = 0011001000 (see <a href="#">Table 17. Look-up table for sink PDO current configuration</a> ) |
| SHIFT_VBUS_HL1        | Coefficient to shift up nominal V <sub>BUS</sub> high voltage limit applicable to 5 V and SNK_PDO1 voltage  | 1010b<br>(10%)        | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBH1}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBH1</sub> = 10%                                                                                                                                    |
| SHIFT_VBUS_LL1        | Coefficient to shift down nominal V <sub>BUS</sub> low voltage limit applicable to 5 V and SNK_PDO1 voltage | 1111b<br>(15%)        | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBL1}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBL1</sub> = 15%                                                                                                                                    |
| SHIFT_VBUS_HL2        | Coefficient to shift up nominal V <sub>BUS</sub> high voltage limit applicable to SNK_PDO2 voltage          | 0101b<br>(5%)         | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBH2}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBH2</sub> = 5%                                                                                                                                     |
| SHIFT_VBUS_LL2        | Coefficient to shift down nominal V <sub>BUS</sub> applicable to SNK_PDO2 voltage                           | 1111b<br>(15%)        | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBL2}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBL2</sub> = 15%                                                                                                                                    |
| SHIFT_VBUS_HL3        | Coefficient to shift up nominal V <sub>BUS</sub> high voltage limit applicable to SNK_PDO3 voltage          | 0101b<br>(5%)         | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBH3}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBH3</sub> = 5%                                                                                                                                     |
| SHIFT_VBUS_LL3        | Coefficient to shift down nominal V <sub>BUS</sub> low voltage limit applicable to SNK_PDO3 voltage         | 1111b<br>(15%)        | 0001b to 1111b     | $1\% \leq V_{\text{SHUSBL3}} \leq 15\%$ of V <sub>BUS</sub> by increment of 1%<br>Default V <sub>SHUSBL3</sub> = 15%                                                                                                                                    |
| VBUS_DISCH_TIME_TO_0V | Coefficient used to compute V <sub>BUS</sub> discharge time to 0V                                           | 1001b<br>(9)          | 0001b to 1111b     | $1 \leq T_{\text{DISPAR0V}} \leq 15$ by increment of 1<br>Unit discharge time: 84 ms (typ.)<br>Default coefficient T <sub>DISPAR0V</sub> = 9, discharge time T <sub>DISUSB0V</sub> = 756 ms                                                             |

| Parameter name         | Parameter description                                                                                               | Reset value (default) | Value          | Description                                                                                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VBUS_DISCH_TIME_TO_PDO | Coefficient used to compute $V_{BUS}$ discharge time when transitioning to lower PDO voltage                        | 1100b (12)            | 0001b to 1111b | $1 \leq T_{DISPARPDO} \leq 15$ by increment of 1<br>Unit discharge time: 24 ms (typ.)<br>Default coefficient $T_{DISPARPDO} = 12$ , discharge time $T_{DISUSBPDO} = 288$ ms |
| VBUS_DISCH_DISABLE     | $V_{BUS}$ discharge deactivation on VBUS_VS_DISCH and DISCH pins                                                    | 0b                    | 0b             | $V_{BUS}$ discharge enabled                                                                                                                                                 |
|                        |                                                                                                                     |                       | 1b             | $V_{BUS}$ discharge disabled                                                                                                                                                |
| USB_COMM_CAPABLE       | USB 2.0 or 3.x data communication capability by sink system                                                         | 0b                    | 0b             | Sink does not support data communication                                                                                                                                    |
|                        |                                                                                                                     |                       | 1b             | Sink supports data communication                                                                                                                                            |
| SNK_UNCONS_POWER       | Unconstrained Power bit setting in capabilities message sent by the sink                                            | 0b                    | 0b             | No external source of power                                                                                                                                                 |
|                        |                                                                                                                     |                       | 1b             | An external source of power is available and is sufficient to adequately power the system while charging external devices                                                   |
| REQ_SRC_CURRENT        | In case of match, selects which operating current from the sink or the source is to be requested in the RDO message | 0b                    | 0b             | Request I(SNK_PDO) as operating current in RDO message                                                                                                                      |
|                        |                                                                                                                     |                       | 1b             | Request I(SRC_PDO) as operating current in RDO message                                                                                                                      |
| POWER_OK_CFG[1:0]      | Selects POWER_OK pins configuration<br>(see <a href="#">Section 2.3.8 POWER_OK2 / POWER_OK3</a> )                   | 10b                   | 00b            | Configuration 1                                                                                                                                                             |
|                        |                                                                                                                     |                       | 01b            | Not applicable                                                                                                                                                              |
|                        |                                                                                                                     |                       | 10b            | Configuration 2 (default)                                                                                                                                                   |
|                        |                                                                                                                     |                       | 11b            | Configuration 3                                                                                                                                                             |
| POWER_ONLY_ABOVE_5V    | Selects VBUS_EN_SNK pin configuration<br>(see <a href="#">Section 2.3.10 VBUS_EN_SNK</a> )                          | 0b                    | 0b             | VBUS_EN_SNK pin enabled when source attached whatever VBUS voltage (5 V or any PDO voltage)                                                                                 |
|                        |                                                                                                                     |                       | 1b             | VBUS_EN_SNK pin enabled only when source attached and VBUS voltage negotiated to PDO2 or PDO3 voltage                                                                       |
| GPIO_CFG[1:0]          | Selects GPIO pin configuration<br>(see <a href="#">Section 2.3.9 GPIO</a> )                                         | 01b                   | 00b            | SW_CTRL_GPIO                                                                                                                                                                |
|                        |                                                                                                                     |                       | 01b            | ERROR_RECOVERY                                                                                                                                                              |
|                        |                                                                                                                     |                       | 10b            | DEBUG                                                                                                                                                                       |
|                        |                                                                                                                     |                       | 11b            | SINK_POWER                                                                                                                                                                  |

**Table 17. Look-up table for sink PDO current configuration**

| Parameter name     | Parameter value | PDO current value                                                                                                                    | Description                |
|--------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| I_SNK_PDO_LUT[3:0] | 0000b           | $0.01 \leq 0.01 \cdot I\_SNK\_PDO\_FLEX[9:0] \leq 5$ by steps of 10 mA<br>Default I_SNK_PDO_FLEX[9:0] = 0011001000<br>(0.01*200=2 A) | Flexible current value     |
|                    | 0001b           | 0.50 A                                                                                                                               | Pre-defined current values |
|                    | 0010b           | 0.75 A                                                                                                                               |                            |
|                    | 0011b           | 1.00 A                                                                                                                               |                            |
|                    | 0100b           | 1.25 A                                                                                                                               |                            |
|                    | 0101b           | 1.50 A                                                                                                                               |                            |
|                    | 0110b           | 1.75 A                                                                                                                               |                            |
|                    | 0111b           | 2.00 A                                                                                                                               |                            |
|                    | 1000b           | 2.25 A                                                                                                                               |                            |
|                    | 1001b           | 2.50 A                                                                                                                               |                            |
|                    | 1010b           | 2.75 A                                                                                                                               |                            |
|                    | 1011b           | 3.00 A                                                                                                                               |                            |
|                    | 1100b           | 3.50 A                                                                                                                               |                            |
|                    | 1101b           | 4.00 A                                                                                                                               |                            |
|                    | 1110b           | 4.50 A                                                                                                                               |                            |
|                    | 1111b           | 5.00 A                                                                                                                               |                            |

**Table 18. STUSB4500 default sink PDO programming**

| Sink PDO # | Type             | Priority | PDO value    | V <sub>BUS</sub> monitoring<br>Low voltage limit<br>V <sub>MONUSBL</sub> | V <sub>BUS</sub> monitoring<br>High voltage limit<br>V <sub>MONUSBH</sub> |
|------------|------------------|----------|--------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| PDO1       | Fixed voltage    | Low      | 5 V / 1.5 A  | 3.3 V (detachment)<br>-20 % (attachment)                                 | +15 %                                                                     |
| PDO2       | Flexible voltage | Medium   | 15 V / 1.5 A | -20 %                                                                    | +10 %                                                                     |
| PDO3       | Flexible voltage | High     | 20 V / 1.0 A | -20 %                                                                    | +10 %                                                                     |

See [Section 7.3 Electrical and timing characteristics](#) for parameters related to V<sub>BUS</sub>.

## 6 Application

The sections below are not part of the ST product specification. They are intended to give a generic application overview to be used by the customer as a starting point for further implementations and customizations. ST does not warrant compliance with customer specifications. Full system implementation and validation are under the customer's responsibility.

### 6.1 General information

#### 6.1.1 Power supplies

The STUSB4500 can be supplied in three different ways depending on the targeted application:

- Through the VDD pin only for applications powered by  $V_{BUS}$  that operate with dead battery mode support
- Through the VSYS pin only for AC powered applications with a system power supply delivering 3.3 V or 5 V
- Through the VDD and VSYS pins either for applications powered by a battery with dead battery mode support or for applications powered by  $V_{BUS}$  with a system power supply delivering 3.3 V or 5 V. When both VDD and VSYS power supplies are present, the low power supply VSYS is selected when VSYS voltage is above 3.1 V. Otherwise VDD is selected

#### 6.1.2 Powering a system under high charging profile only

The STUSB4500 allows the  $V_{BUS}$  power path to be enabled through VBUS\_EN\_SNK pin only when a high power charging profile above 5 V has been negotiated (PDO2 or PDO3) with the source (see [Section 3.3 Auto-run mode](#)).

This feature can be turned on thanks to NVM bit POWER\_ONLY\_ABOVE\_5V (see [Section 5 Start-up configuration](#)). When the bit value is set to logic level high, the VBUS\_EN\_SNK pin is asserted only when a PDO2 or PDO3 explicit contract is established with the source (see [Section 2.3.10 VBUS\\_EN\\_SNK](#)).

In case of mismatch, the  $V_{BUS}$  power path remains open while the source provides 5 V on the USB Type-C receptacle. The source and the sink stay electrically connected through the CC pins. Thus, when the source is able later to provide power capabilities corresponding to those expected by the sink, a new negotiation is again possible upon source request. If the PDO negotiation succeeds, the VBUS\_EN\_SNK pin is asserted which allows the system to be powered at the negotiated PDO profile.

This feature is suited for sink devices requiring high power charging profile above 5 V to be fully operational.

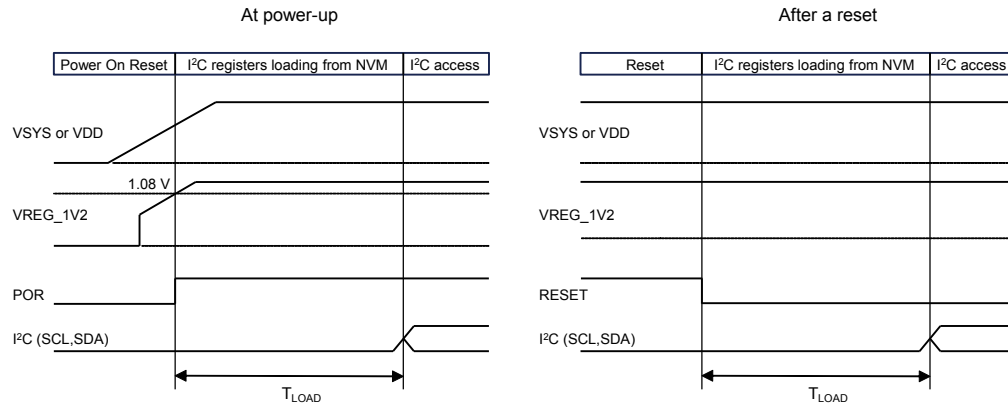
#### 6.1.3 Connection to MCU or application processor

The STUSB4500 runs as a standalone USB PD sink controller. The connection to an MCU or an application processor is optional. However, an I<sup>2</sup>C interface with an interrupt allows simple connection to most of MCU and SOC of the market.

When a connection through the I<sup>2</sup>C interface is implemented, it provides extensive functionality during the system operation. For instance, it may be used to:

1. Define the port configuration during system boot (in case the NVM parameters are not customized during manufacturing)
2. Provide a diagnostic of the Type-C connection in real time

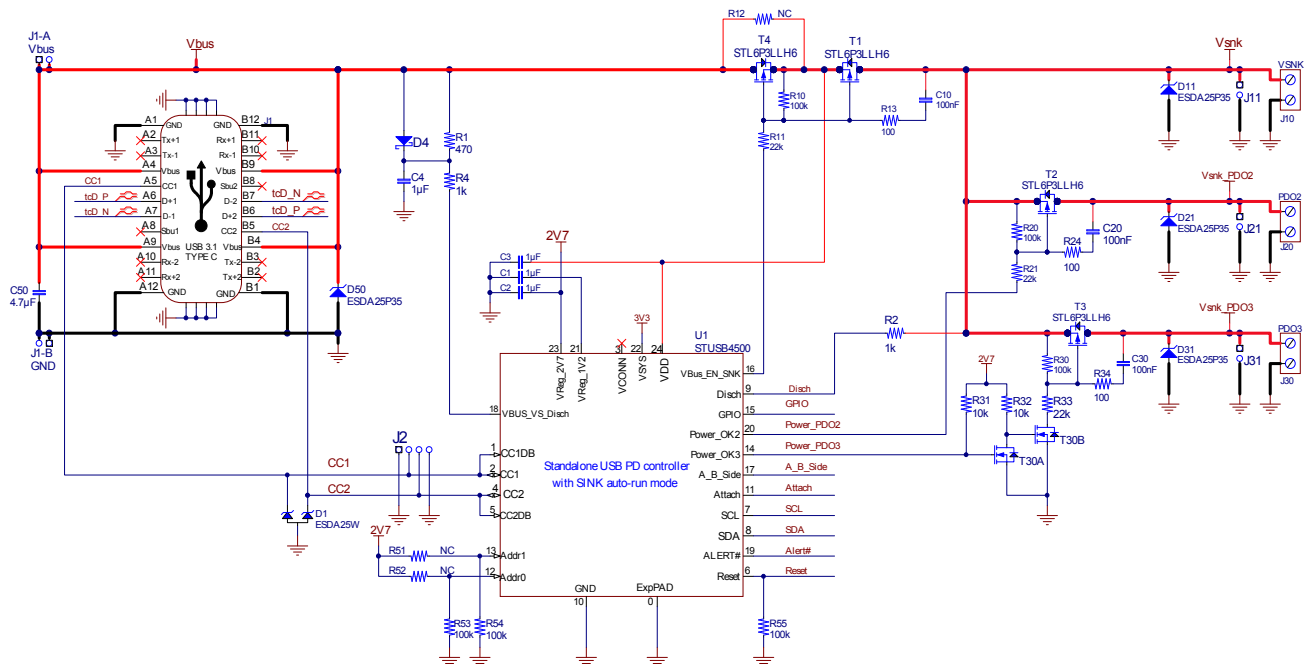
At power-up or after a reset, the first software access to the I<sup>2</sup>C registers of the STUSB4500 can be done only after  $T_{LOAD}$  as shown in the figure below.  $T_{LOAD}$  corresponds to the time required to initialize the I<sup>2</sup>C registers with the default values from the embedded NVM. At power-up, the loading phase starts when the voltage level on the VREG\_1V2 output pin of the 1.2 V internal regulator reaches 1.08 V to release the internal POR signal. After a reset, the loading phase starts when the signal on the RESET pin is released.

**Figure 6. I<sup>2</sup>C register initialization sequence at power-up or after a reset**


## 6.2 Typical applications

### 6.2.1 Sink USB PD application with dual high power charging paths

#### 6.2.1.1 Application schematic

**Figure 7. Implementation example with POWER\_OK pins configuration #2 (default)**


### 6.2.1.2 VBUS power path activation

**Table 19.** VBUS power path activation versus negotiated PDO contract

| Negotiated<br>PDO contract                  | Transistors T4/T1<br>(VBUS_EN_SNK) | Transistor T2<br>(POWER_OK2) | Transistor T3<br>(POWER_OK3) |
|---------------------------------------------|------------------------------------|------------------------------|------------------------------|
| Implicit 5 V USB Type-C<br>or PDO1 contract | ON                                 | OFF                          | OFF                          |
| PDO2 contract                               | ON                                 | ON                           | OFF                          |
| PDO3 contract                               | ON                                 | OFF                          | ON                           |

## 7 Electrical characteristics

### 7.1 Absolute maximum ratings

All voltages are referenced to GND.

**Table 20. Absolute maximum ratings**

| Symbol                                                                                                                                                                                                                    | Parameter                             | Value      | Unit |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------|------|
| V <sub>DD</sub>                                                                                                                                                                                                           | Supply voltage on VDD pin             | 28         | V    |
| V <sub>SYS</sub>                                                                                                                                                                                                          | Supply voltage on VSYS pin            | 6          |      |
| V <sub>CC1</sub> , V <sub>CC2</sub><br>V <sub>CC1DB</sub> , V <sub>CC2DB</sub>                                                                                                                                            | High voltage on CC pins               | 22         |      |
| V <sub>VBUS_EN_SNK</sub><br>V <sub>VBUS_VS_DISCH</sub><br>V <sub>DISCH</sub><br>V <sub>POWER_OK2</sub>                                                                                                                    | High voltage on V <sub>BUS</sub> pins | 28         |      |
| V <sub>SCL</sub> , V <sub>SDA</sub><br>V <sub>ALERT</sub><br>V <sub>RESET</sub><br>V <sub>ATTACH</sub><br>V <sub>A_B_SIDE</sub><br>V <sub>POWER_OK3</sub><br>V <sub>GPIO</sub><br>V <sub>ADDR0</sub> , V <sub>ADDR1</sub> | Operating voltage on I/O pins         | -0.3 to 6  |      |
| T <sub>STG</sub>                                                                                                                                                                                                          | Storage temperature                   | -55 to 150 |      |
| T <sub>J</sub>                                                                                                                                                                                                            | Maximum junction temperature          | 145        |      |
| ESD                                                                                                                                                                                                                       | HBM                                   | 3          |      |
|                                                                                                                                                                                                                           | CDM                                   | 1.5        |      |

## 7.2 Operating conditions

**Table 21. Operating conditions**

| Symbol                                                                                                                                                                                                                    | Parameter                     | Value      | Unit |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------|------|
| V <sub>DD</sub>                                                                                                                                                                                                           | Supply voltage on VDD pin     | 4.1 to 22  | V    |
| V <sub>SYS</sub>                                                                                                                                                                                                          | Supply voltage on VSYS pin    | 3.0 to 5.5 |      |
| V <sub>CC1</sub> , V <sub>CC2</sub><br>V <sub>CC1DB</sub> , V <sub>CC2DB</sub>                                                                                                                                            | CC pins                       | 0 to 5.5   |      |
| V <sub>VBUS_EN_SNK</sub><br>V <sub>VBUS_VS_DISCH</sub><br>V <sub>DISCH</sub><br>V <sub>POWER_OK2</sub>                                                                                                                    | High voltage pins             | 0 to 22    |      |
| V <sub>SCL</sub> , V <sub>SDA</sub><br>V <sub>ALERT</sub><br>V <sub>RESET</sub><br>V <sub>ATTACH</sub><br>V <sub>A_B_SIDE</sub><br>V <sub>POWER_OK3</sub><br>V <sub>GPIO</sub><br>V <sub>ADDR0</sub> , V <sub>ADDR1</sub> | Operating voltage on I/O pins | 0 to 4.5   |      |
| T <sub>A</sub>                                                                                                                                                                                                            | Operating temperature         | -40 to 105 | °C   |

## 7.3 Electrical and timing characteristics

Unless otherwise specified: V<sub>DD</sub> = 5 V, T<sub>A</sub> = 25 °C, all voltages are referenced to GND.

**Table 22. Electrical characteristics**

| Symbol                  | Parameter                                        | Conditions                                                                                         | Min. | Typ. | Max. | Unit |
|-------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------|------|------|------|------|
| I <sub>DD</sub> (SNK)   | Current consumption                              | Device idle as a SINK<br>(not connected, no communication)<br>V <sub>SYS</sub> @ 3.3 V             | 115  | 140  | 165  | μA   |
|                         |                                                  | V <sub>DD</sub> @ 5.0 V                                                                            | 110  | 160  | 210  | μA   |
| T <sub>LOAD</sub>       | I <sup>2</sup> C registers loading time from NVM | At power-up or after a reset                                                                       |      |      | 30   | ms   |
| <b>CC1 and CC2 pins</b> |                                                  |                                                                                                    |      |      |      |      |
| R <sub>d</sub>          | CC pull-down resistors                           | -40 °C < T <sub>A</sub> < +105 °C                                                                  | -10% | 5.1  | +10% | kΩ   |
| R <sub>INCC</sub>       | CC input impedance                               | Terminations off                                                                                   | 200  |      |      | kΩ   |
| V <sub>TH0.2</sub>      | Detection threshold 1                            | Min. I <sub>P-USB</sub> detection by sink on R <sub>d</sub> ,<br>min CC voltage for connected sink | 0.15 | 0.20 | 0.25 | V    |
| V <sub>TH0.66</sub>     | Detection threshold 2                            | Min. I <sub>P_1.5</sub> detection by sink on R <sub>d</sub>                                        | 0.61 | 0.66 | 0.71 | V    |

| Symbol                                          | Parameter                                                                  | Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Min.                          | Typ.                                         | Max.                          | Unit |
|-------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------|-------------------------------|------|
| V <sub>TH1.23</sub>                             | Detection threshold 3                                                      | Min. I <sub>P_3.0</sub> detection by sink on Rd                                                                                                                                                                                                                                                                                                                                                                                                                         | 1.16                          | 1.23                                         | 1.31                          | V    |
| V <sub>TH2.6</sub>                              | Detection threshold 4                                                      | Max. CC voltage for connected sink                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2.45                          | 2.60                                         | 2.75                          | V    |
| V <sub>OVP</sub>                                | Overvoltage protection on CC pins                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 5.82                          | 6                                            | 6.18                          | V    |
| <b>VBUS_VS_DISCH pin monitoring and driving</b> |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |                                              |                               |      |
| V <sub>THUSB</sub>                              | V <sub>BUS</sub> disconnection threshold (5 V USB Type-C or PDO1 selected) | V <sub>SYS</sub> = 3.0 to 5.5 V                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3.2                           | 3.3                                          | 3.4                           | V    |
| V <sub>TH0V</sub>                               | V <sub>BUS</sub> safe 0 V threshold (vSafe0V)                              | V <sub>SYS</sub> = 3.0 to 5.5 V                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.5                           | 0.6                                          | 0.7                           | V    |
| I <sub>DISUSB</sub>                             | V <sub>BUS</sub> discharge current                                         | Through external resistor connected to VBUS_VS_DISCH pin                                                                                                                                                                                                                                                                                                                                                                                                                |                               |                                              | 50                            | mA   |
| T <sub>DISUSB0V</sub>                           | V <sub>BUS</sub> discharge time to 0 V                                     | At detachment, during error recovery state or hard reset,<br>Coefficient T <sub>DISPAR0V</sub> programmable by NVM,<br>Default T <sub>DISPAR0V</sub> = 9, T <sub>DISUSB0V</sub> = 756 ms                                                                                                                                                                                                                                                                                | 70<br>*T <sub>DISPAR0V</sub>  | 84<br>*T <sub>DISPAR0V</sub>                 | 100<br>*T <sub>DISPAR0V</sub> | ms   |
| T <sub>DISUSBPDO</sub>                          | V <sub>BUS</sub> transition discharge time to new PDO                      | At transition to a lower PDO voltage,<br>Coefficient T <sub>DISPARPDO</sub> programmable by NVM,<br>Default T <sub>DISPARPDO</sub> = 12, T <sub>DISUSBPDO</sub> = 288 ms                                                                                                                                                                                                                                                                                                | 20<br>*T <sub>DISPARPDO</sub> | 24<br>*T <sub>DISPARPDO</sub>                | 28<br>*T <sub>DISPARPDO</sub> | ms   |
| V <sub>MONUSBH</sub>                            | V <sub>BUS</sub> monitoring high voltage limit                             | V <sub>BUS</sub> can be 5 V <sub>USB</sub> Type-C voltage or any PDO voltage,<br>V <sub>BUS</sub> +5% is nominal high voltage limit,<br>Shift coefficient V <sub>SHUSBH</sub> is programmable by NVM from 1% to 15% of V <sub>BUS</sub> by step of 1%,<br>Default<br>V <sub>SHUSBH1</sub> = 10%, V <sub>MONUSBH1</sub> = V <sub>BUS</sub> +15% (5 V USB Type-C or PDO1),<br>V <sub>SHUSBH2/3</sub> = 5%, V <sub>MONUSBH2/3</sub> = V <sub>BUS</sub> +10% (PDO2 or PDO3) |                               | V <sub>BUS</sub> +5%<br>+V <sub>SHUSBH</sub> |                               | V    |
| V <sub>MONUSBL</sub>                            | V <sub>BUS</sub> monitoring low voltage limit                              | V <sub>BUS</sub> can be 5 V <sub>USB</sub> Type-C voltage or any PDO voltage,<br>V <sub>BUS</sub> -5% is nominal low voltage limit,<br>Shift coefficient V <sub>SHUSBL</sub> is programmable by NVM from 1% to 15% of V <sub>BUS</sub> by step of 1%,<br>Default<br>V <sub>SHUSBL1/2/3</sub> = 15%, V <sub>MONUSBL1/2/3</sub> = V <sub>BUS</sub> -20% (5 V USB Type-C or any PDO)                                                                                       |                               | V <sub>BUS</sub> -5%<br>-V <sub>SHUSBL</sub> |                               | V    |
| <b>DISCH pin driving</b>                        |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                               |                                              |                               |      |
| I <sub>DISPWR</sub>                             | Power system discharge current                                             | Through external resistor connected to DISCH pin                                                                                                                                                                                                                                                                                                                                                                                                                        |                               |                                              | 500                           | mA   |

| Symbol                                                                                                | Parameter                | Conditions             | Min. | Typ. | Max. | Unit |
|-------------------------------------------------------------------------------------------------------|--------------------------|------------------------|------|------|------|------|
| <b>Digital input/output (SCL, SDA, ALERT, RESET, ATTACH, A_B_SIDE, POWER_OK3, GPIO, ADDR0, ADDR1)</b> |                          |                        |      |      |      |      |
| V <sub>IH</sub>                                                                                       | High level input voltage |                        | 1.2  |      |      | V    |
| V <sub>IL</sub>                                                                                       | Low level input voltage  |                        |      |      | 0.35 | V    |
| V <sub>OL</sub>                                                                                       | Low level output voltage | I <sub>oh</sub> = 3 mA |      |      | 0.4  | V    |
| <b>20 V open drain outputs (VBUS_EN_SNK, DISCH, POWER_OK2)</b>                                        |                          |                        |      |      |      |      |
| V <sub>OL</sub>                                                                                       | Low level output voltage | I <sub>oh</sub> = 3 mA |      |      | 0.4  | V    |

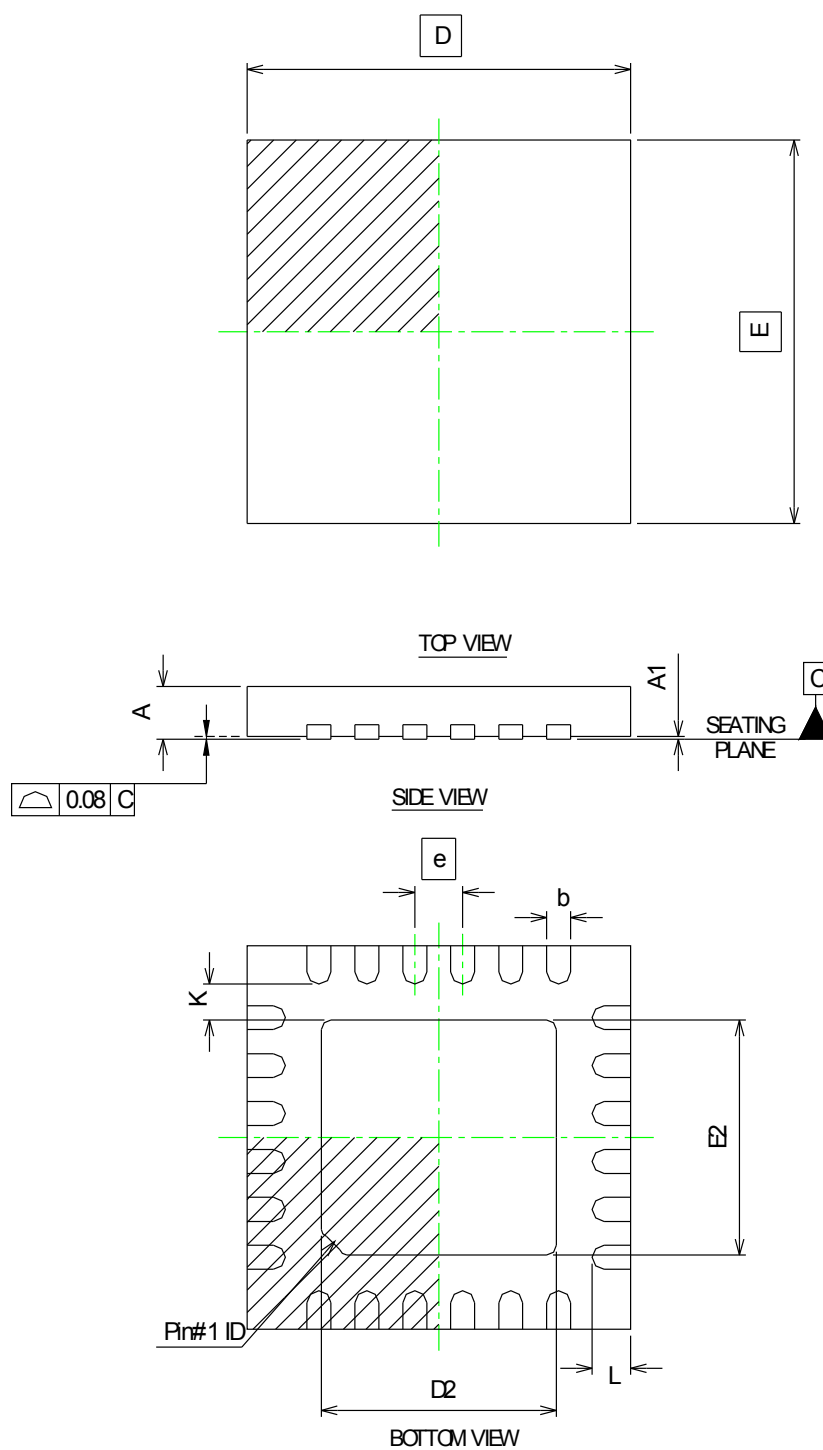
## 8 Package information

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In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: [www.st.com](http://www.st.com). ECOPACK® is an ST trademark.

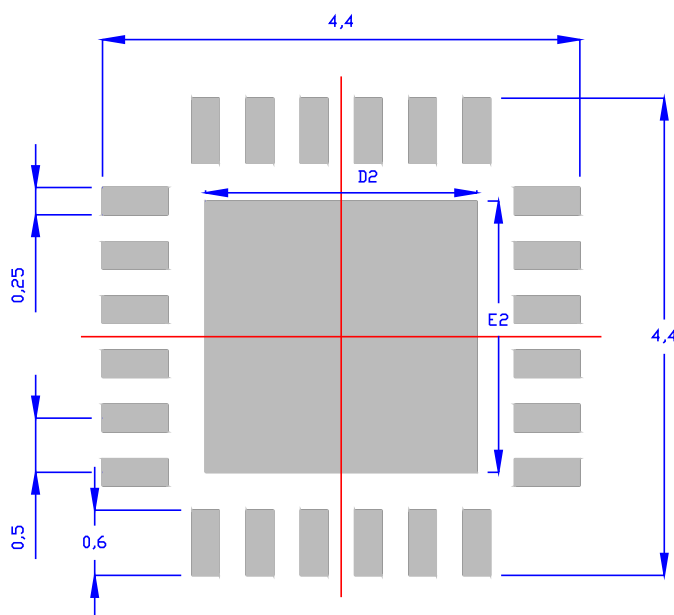
## 8.1 QFN24 EP 4x4 mm package information

Figure 8. QFN24 EP 4x4 mm package information



**Table 23. QFN24 EP 4x4 mm package mechanical data**

| Ref. | Millimeters |      |      | Inches |        |       |
|------|-------------|------|------|--------|--------|-------|
|      | Min.        | Typ  | Max. | Min.   | Typ.   | Max.  |
| A    | 0.80        | 0.90 | 1.00 | 0.031  | 0.035  | 0.039 |
| A1   | 0.00        | 0.02 | 0.05 | 0.000  | 0.001  | 0.002 |
| b    | 0.18        | 0.25 | 0.30 | 0.007  | 0.0010 | 0.012 |
| D    | 3.95        | 4.00 | 4.05 | 0.156  | 0.157  | 0.159 |
| D2   | 2.55        | 2.70 | 2.80 | 0.100  | 0.106  | 0.110 |
| E    | 3.95        | 4.00 | 4.05 | 0.156  | 0.157  | 0.159 |
| E2   | 2.55        | 2.70 | 2.80 | 0.100  | 0.106  | 0.110 |
| e    | 0.45        | 0.50 | 0.55 | 0.018  | 0.020  | 0.022 |
| K    | 0.15        | -    | -    | 0.006  | -      | -     |
| L    | 0.30        | 0.40 | 0.50 | 0.012  | 0.016  | 0.020 |

**Figure 9. QFN24 EP 4x4 mm recommended footprint**


## 8.2 Thermal information

**Table 24. Thermal information**

| Symbol          | Parameter                              | Value | Unit |
|-----------------|----------------------------------------|-------|------|
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance | 37    | °C/W |
| $R_{\theta JC}$ | Junction-to-case thermal resistance    | 5     |      |

## 9 Terms and abbreviations

**Table 25. List of terms and abbreviations**

| Term           | Description                                                                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Accessory mode | Debug accessory mode. It is defined by the presence of pull-up resistors $R_p/R_p$ on CC1/CC2 pins in sink power role.                                                                                                                                             |
| DFP            | Downstream facing port, specifically associated with the flow of data in a USB connection. Typically the ports on a HOST or the ports on a hub to which devices are connected. In its initial state, the DFP sources $V_{BUS}$ and $V_{CONN}$ , and supports data. |
| DRP            | Dual-role port. A port that can operate as either a source or a sink. The port role may be changed dynamically.                                                                                                                                                    |
| Sink           | Port asserting $R_d$ on the CC pins and consuming power from the $V_{BUS}$ ; most commonly a device.                                                                                                                                                               |
| Source         | Port asserting $R_p$ on the CC pins and providing power over the $V_{BUS}$ ; most commonly a host or hub DFP.                                                                                                                                                      |
| UFP            | Upstream facing port, specifically associated with the flow of data in a USB connection. The port on a device or a hub that connects to a host or the DFP of a hub. In its initial state, the UFP sinks $V_{BUS}$ and supports data.                               |

## Revision history

**Table 26. Document revision history**

| Date        | Revision | Changes          |
|-------------|----------|------------------|
| 06-Apr-2018 | 1        | Initial release. |

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